

ASSIGNMENT #5 - NEURAL NETWORKS

CONTEXT

This assignment is an opportunity to demonstrate your knowledge and practice solving problems about attention. This assignment contains an implementation component (i.e. you will be asked to write code to solve one or more problems) and an experiment component (i.e. you will be asked to experiment with your code and report results).

LOGISTICS

Assignment due date: 2026-04-03

Assignments are to be submitted electronically through Brightspace. It is your responsibility to ensure that your assignment is submitted properly and that all the files for the assignment are included. Copying of assignments is NOT allowed. High-level discussion of assignment work with others is acceptable, but each individual or small group is expected to do the work themselves.

IMPLEMENTATION PART

Programming language: Python 3

For all parts of the implementation, you may use the Python Standard Library (<https://docs.python.org/3/library/>) and the following packages (and any packages they depend on): NumPy, Pandas, PyTorch, scikit-learn, SciPy. Unless explicitly indicated below or explicitly approved by the instructor, you may not use any additional packages.

You must implement your code yourself; do not copy-and-paste code from other sources. Please ensure your implementation follows the specifications provided; it may be tested on several different test cases for correctness. Please make sure your code is readable; it may also be manually assessed for correctness. You do not need to prove correctness of your implementation.

You must submit your implementation as a single file named “assignment5.py” with functions as described below (you may have other variables/functions/classes in your file). Attached is skeleton code indicating the format your implementation should take.

EXPERIMENTS PART

You must submit your report on experimental results as a single file named “assignment5.pdf”.

IMPLEMENTATION PART

For the implementation, we will use a financial news dataset. This dataset consists of financial news headlines with corresponding sentiment labels (negative, neutral, positive). This dataset is publicly available on Kaggle: <https://www.kaggle.com/datasets/ankurzing/sentiment-analysis-for-financial-news>. For more information on the dataset, you may refer to the following paper.

Malo P, Sinha A, Korhonen P, Wallenius J, Takala P. Good debt or bad debt: Detecting semantic orientations in economic texts. *Journal of the Association for Information Science and Technology*. 2014 Apr;65(4):782-96.

Note that the quality of the headlines within this dataset has not been verified, and many headlines contain non-alphanumeric characters and non-standard whitespace. You should ensure your implementation is robust to any quality issues with the dataset.

The CSV file contains two columns: the first column indicates the sentiment and the second column indicates the headline.

Use the following one-hot encoding scheme to represent class labels as one-hot vectors.

negative $\rightarrow [1, 0, 0]$

neutral $\rightarrow [0, 1, 0]$

positive $\rightarrow [0, 0, 1]$

QUESTION 1 [10 MARKS]

Create a PyTorch dataset class for this dataset (i.e. a class that subclasses `torch.utils.data.Dataset`). The dataset should be usable for training or testing a neural network in PyTorch. The data generator should perform appropriate pre-processing on the text (e.g. padding/cropping, tokenization, cleaning, etc.). You may also consider adding data augmentation in the data generator.

The data generator class must be called “FinancialNewsDataset”, and it should have three member functions: “__init__”, “__len__”, “__getitem__” (you may also use other helper functions).

The function “__init__” is a constructor that should take two input arguments (in addition to “self”). The first input argument is the full path to the directory containing the dataset. The second input argument is a list of rows from the dataset that should be used (for example, if this argument is [1, 200, 3000], your data generator should only use the headlines in rows 1, 200, and 3000 of the file).

The function “__len__” should take zero input arguments (in addition to “self”). It should return the total number of samples in the subset of dataset.

The function “__getitem__” should take one input argument (in addition to “self”). The first input argument is a sample index. The function should return two values: (1) a sample of data (i.e. one headline) and (2) the label associated with the sample (i.e. the headline’s sentiment).

QUESTION 2 [10 MARKS]

Write a function that creates, trains, and evaluates a recurrent neural network to predict the sentiment of a financial news headline. You may consider using an embedding layer to pre-process the data prior to the recurrent layer(s).

The function must be called “financial_news_rnn_model”.

The function should take three input arguments: (1) the full file path to the data, (2) a list of rows to use from the file for training, and (3) a list of rows to use from the file for validation.

The function should return three values: (1) a PyTorch module object (subclassing "torch.nn.Module") that is a recurrent neural network for this dataset, (2) performance of the model on a training set, and (3) the performance of the model on a validation set.

(Hint: use the data generator you wrote)

QUESTION 3 [10 MARKS]

Write a function that creates, trains, and evaluates an attention-based neural network to predict the sentiment of a financial news headline. You may consider using an embedding layer to pre-process the data prior to the attention-based layer(s).

The function must be called “financial_news_attention_model”.

The function should take three input arguments: (1) the full file path to the data, (2) a list of rows to use from the file for training, and (3) a list of rows to use from the file for validation.

The function should return three values: (1) a PyTorch module object (subclassing "torch.nn.Module") that is an attention-based neural network for this dataset, (2) performance of the model on a training set, and (3) the performance of the model on a validation set.

(Hint: use the data generator you wrote)

EXPERIMENTS PART

QUESTION 4 [10 MARKS]

Conduct a series of experiments on the attention-based neural network you implemented for this. For each of these experiments, plot the performance on the training set and validation set as a function of the parameters.

- a) Experiment on the size of the attention layers.
- b) Experiment on the number of attention layers.
- c) Experiment on the number of epochs of training.
- d) Experiment on the value of the learning rate.

Report the parameters and performance of the model with the best performance. Comment on any trends you observed in the results of your experiments.

- e) Using the best performing model you found, compute performance on the held-out test set.

(Hint: use the data generator you wrote)